Math 54 Discussion Worksheet

Name: ____ ID: __________________________

Problem 1

Let $\{v_1, v_2, v_3, v_4\}$ be a linearly dependent spanning set for a vector space V. Show that each $w \in V$ can expressed in more than one way as a linear combination of the v_i's. *Hint: Start by writing* $w = a_1v_1 + a_2v_2 + a_3v_3 + a_4v_4$ *and use the linear dependence of the* v_i *to find another expression for* w.

Problem 2

Let $p_1(t) = 1 + t^2,\ p_2(t) = t - 3t^2,\ p_3(t) = 1 + t - 3t^2$.

a) Use coordinate vectors to show that these form a basis for $\mathbb{P}_2$.

b) Let $\mathcal{B} = \{p_1, p_2, p_3\}$ be a basis for $\mathbb{P}_2$. Find $q \in \mathbb{P}_2$ given that

$$[q]_{\mathcal{B}} = \begin{bmatrix} -1 \\ 1 \\ 2 \end{bmatrix}$$

Problem 3

Let A be an $m \times n$ matrix. Prove the Rank-Nullity Theorem:

$$\operatorname{rank}(A) + \operatorname{nullity}(A) = n.$$

Hint: Count the pivot columns of A.